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Garbage-collection in log-based block devices with snapshots

Abstract

A method for garbage-collection includes obtaining a request to compact a plurality of log files of a log-structured volume. Each log file includes fresh block runs in use and stale block runs no longer in use. The log-structured volume includes a plurality of snapshots. For each respective snapshot, the method includes determining, using a plurality of interval maps, the fresh block runs of the plurality of log files used by the respective snapshot. For each respective log file, the method includes writing the fresh block runs of the respective log file to a respective compacted log file and generating a respective per-log diff file. The method includes, for each respective snapshot, generating a respective checkpoint based on respective per-log diff files and deleting each respective log file of the plurality of log files.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates to garbage collection in log-based block devices with snapshots.

BACKGROUND

(2) Log-based storage devices store data and metadata in one or more logs. That is, data is written to disk sequentially rather than scanning blocks across the disk. Because data is written sequentially, log-based storage devices may transform groups of small, synchronous writes into a few large, asynchronous sequential write operations. This can greatly increase write bandwidth by decreasing seek overhead. However, log-structured volumes and file systems must include careful management of free space. As the log begins to fill, garbage collection must be undertaken to free space in the log by deleting data no longer in use.

SUMMARY

(3) One aspect of the disclosure provides a computer-implemented method for garbage collection of log-based block device that, when executed by data processing hardware, causes the data processing hardware to perform operations. The operations include obtaining a request to compact a plurality of log files of a log-structured volume. The log-structured volume includes a plurality of snapshots. Each snapshot of the plurality of snapshots is representative of a state of the log-structured volume at a different point in time. For each respective snapshot of the plurality of snapshots of the log-structured volume, the operations include determining fresh block runs of the plurality of log files used by the respective snapshot. For each respective log file of the plurality of log files, the operations include writing the fresh block runs of the respective log file to a respective compacted log file and generating a respective per-log diff file mapping a location of the written fresh block runs at the respective log file to a location of the written fresh block runs at the

respective compacted log file. For each respective snapshot of the plurality of snapshots of the log-structured volume, the operations include generating a respective checkpoint based on respective per-log diff files and deleting each respective log file of the plurality of log files.

(4) Implementations of the disclosure may include one or more of the following optional features. In some implementations, determining the fresh block runs includes using a plurality of interval maps where each respective interval map of the plurality of interval maps specifies the fresh block runs of a respective log file for each snapshot of the plurality of snapshots and the operations further include generating, using the plurality of log files and the plurality of snapshots, the plurality of interval maps. In some of these implementations, generating the plurality of interval maps includes scanning, for each snapshot of the plurality of snapshots, each log file of the plurality of log files.

(5) Optionally, each fresh block run and each stale block run includes a tuple comprising a log file name, an offset, and a byte count. The operations further include, in some examples, selecting the plurality of log files from a set of log files based on a ratio of a sum of a size of the fresh block runs of each respective log file over a total size of each respective log file. In some of these examples, the operations further include, for each respective log file of the set of log files, determining the ratio for each respective log file using a respective interval map of the plurality of interval maps.

(6) In some implementations, each log file of the plurality of log files is frozen. Obtaining the request to compact the plurality of log files may include determining a threshold amount of time has passed since a previous compaction. Each respective per-log diff file optionally includes an ordered list of tuples, each tuple representing a respective fresh block run and including an offset, a log file name, and a byte count. In some examples, the operations further include, for each deleted respective log file, deleting the respective per-log diff file. In some examples, each log file of the plurality of log files includes the fresh block runs representing contiguous blocks of data in use by the log-structured volume and stale block runs representing contiguous blocks of data no longer in use by the log-structured volume.

(7) Another aspect of the disclosure provides a system for garbage collection of log-based block device. The system includes data processing hardware and memory hardware in communication with the data processing hardware. The memory hardware stores instructions that when executed on the data processing hardware cause the data processing hardware to perform operations. The operations include obtaining a request to compact a plurality of log files of a log-structured volume. The log-structured volume includes a plurality of snapshots. Each snapshot of the plurality of snapshots is representative of a state of the log-structured volume at a different point in time. For each respective snapshot of the plurality of snapshots of the log-structured volume, the operations include determining fresh block runs of the plurality of log files used by the respective snapshot. For each respective log file of the plurality of log files, the operations include writing the fresh block runs of the respective log file to a respective compacted log file and generating a respective per-log diff file mapping a location of the written fresh block runs at the respective log file to a location of the written fresh block runs at the respective compacted log file. For each respective snapshot of the plurality of snapshots of the log-structured volume, the operations include generating a respective checkpoint based on respective per-log diff files and deleting each respective log file of the plurality of log files.

(8) This aspect may include one or more of the following optional features. In some implementations, the operations further include generating, using the plurality of log files and the plurality of snapshots, the plurality of interval maps. In some of these implementations, generating the plurality of interval maps includes scanning, for each snapshot of the plurality of snapshots, each log file of the plurality of log files.

(9) Optionally, each fresh block run and each stale block run includes a tuple comprising a log file name, an offset, and a byte count. The operations further include, in some examples, selecting the plurality of log files from a set of log files based on a ratio of a sum of a size of the fresh block runs

of each respective log file over a total size of each respective log file. In some of these examples, the operations further include, for each respective log file of the set of log files, determining the ratio for each respective log file using a respective interval map of the plurality of interval maps. (10) In some implementations, each log file of the plurality of log files is frozen. Obtaining the request to compact the plurality of log files may include determining a threshold amount of time has passed since a previous compaction. Each respective per-log diff file optionally includes an ordered list of tuples, each tuple representing a respective fresh block run and including an offset, a log file name, and a byte count. In some examples, the operations further include, for each deleted respective log file, deleting the respective per-log diff file. In some examples, each log file of the plurality of log files includes the fresh block runs representing contiguous blocks of data in use by the log-structured volume and stale block runs representing contiguous blocks of data no longer in use by the log-structured volume.

(11) The details of one or more implementations of the disclosure are set forth in the accompanying drawings and the description below. Other aspects, features, and advantages will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic view of an example system for performing garbage collection on log-based block devices with snapshots.

(2) FIGS. 2A-2D are schematic views of exemplary log files during a compaction event.

(3) FIG. 3 is a schematic view of an exemplary snapshot statistics table.

(4) FIG. 4 a flowchart of an example arrangement of operations for a method of performing garbage collection on log-based block devices with snapshots.

(5) FIG. 5 is a schematic view of an example computing device that may be used to implement the systems and methods described herein.

(6) Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

(7) Log-based storage devices and file systems store data and metadata in one or more logs. That is, data is written to disk sequentially rather than scattering blocks across the disk like most conventional file systems. Because data is written sequentially, log-based storage devices may transform groups of small, synchronous writes into a few large, asynchronous sequential write operations. This can greatly increase write bandwidth by decreasing seek overhead. However, log-structured volumes and file systems must include careful management of free space. As the log begins to fill, garbage collection must be undertaken to free space in the log by deleting data no longer in use.

(8) Garbage collection of log-based block devices may be performed by background workers. These workers first load a checkpoint representing a point-in-time view of the device, and proceed to read the entirety of the device (e.g., all of the log files representing the device). These log files are filled with both fresh data that is still in use and stale data that is no longer in use. Next, the worker writes all of the fresh data out to new log files. After this process, the new log files only contain in-use data and the original input log files can be deleted.

(9) The block device may support “snapshots.” A snapshot is a state of the block device at a particular point in time. Snapshots serve many important functions, such as data protection and availability. However, these snapshots increase the complexity of garbage collection. Without performing garbage collection on the log files associated with the snapshots, the amount of space consumed on the device will rapidly increase. However, performing garbage collection on the snapshots is challenging due to the extra storage and processing costs for the worker, especially

when the worker is designed with limited memory as a background process.

(10) A naive implementation for garbage collection for such a system would include opening and reading the log files for each snapshot, writing out a new log file, and generating a new checkpoint for the snapshot. Each checkpoint defines, for each snapshot, the location in the log file of respective block runs (i.e., contiguous ranges of data blocks). However, building checkpoints in this manner requires a significant amount of memory as the worker must track the location of many data blocks. Because it is likely many snapshots “share” the same data (i.e., there may be little changes to the data between snapshots), using such large amounts of memory to generate very similar checkpoints is an inefficient use of resources.

(11) Implementations herein are directed toward a compaction controller of a computing system (e.g., a distributed system) that enables efficient garbage collection or compaction of log-structured volumes (e.g., persistent disks) that employ snapshots. The compaction controller performs a scan to determine, using an interval map, which blocks of the log files are currently fresh or in use. The compaction controller writes all fresh blocks to new log files, and each time a block is moved from a source log file to a new log file, a record is added to a per-log diff file that indicates where the new data is now stored. The compaction controller merges the compacted log files into the system by writing a new checkpoint for each snapshot using the per-log diff files. The compaction controller, instead of tracking all of the new locations of the new data blocks in each of the different checkpoints, generates the per-log diff files per log file. Thus, each log file that the compaction controller compacts has an associated per-log diff file and when building the checkpoints, the compaction controller can go to the same per-log diff file for each snapshot that uses the respective log file, thus leveraging the limited differences in data between snapshots to greatly decrease the amount of resources required to generate the checkpoints.

(12) Referring to FIG. 1, in some implementations, an example data compaction system **100** includes a remote system **140** in communication with one or more user devices **10** via a network **112**. The remote system **140** may be a single computer, multiple computers, or a distributed system (e.g., a cloud environment) having scalable/elastic resources **142** including computing resources **144** (e.g., data processing hardware) and/or storage resources **146** (e.g., memory hardware). A data store (i.e., a remote storage device) may be overlain on the storage resources **146** to allow scalable use of the storage resources **146** by one or more of the clients (e.g., the user device **10**) or the computing resources **144**.

(13) The storage resources **146** include one or more persistent disks **146**, **146P**, **146Pa-n**. Each persistent disk **146** represents a durable network storage device that instances (e.g., virtual machines (VMs)) may access like a physical disk. Each persistent disk **146P** may be distributed across several physical disks (e.g., solid-state drives (SSDs), hard disks, etc.). Each persistent disk **146P** is a block-based log-structured volume. That is, each persistent disk **146P** stores data via one or more log files **210**, **210a-n**. The persistent disks **146P** may be divided into one or more segments of any size (e.g., 32 gigabytes per segment).

(14) As the remote system **140** receives block runs **214**, **214a-n** to write to the persistent disks **146P** (e.g., from user **12**, the remote system **140** itself, a VM, or any other remote entity), the remote system **140** writes the block runs **214** to respective log files **210**. Each block run **214** represents or is used to describe a location of a contiguous range of data blocks **212**, **212a-n** (e.g., **4k** blocks) of the log files **210** and/or persistent disk **146P**. In some examples, each block run **212** is a tuple that includes a log file name (i.e., a reference to which log file **210** the data blocks **212** are at), an offset (i.e., a location with the respective log file **210**), and a byte count (i.e., a length or size of the range of data blocks **212**). In some examples, depending on the file system in use on the persistent disk **146P**, the block runs **214** may represent some or all of a data file. Periodically, or whenever certain conditions are satisfied, the remote system **140** captures a snapshot **156**, **156a-n** of the persistent disk(s) **146P**. Each snapshot **156** captures or represents a state of the persistent disk **146P** at the time of the snapshot **156** (i.e., the state of the block runs **214** at the point in time the snapshot **156** is

taken). As used herein, snapshots **156** may also be referred to as “generations.” Using the snapshots, a user **12** or the remote system **140** may recall the state of the persistent disk **146P** at the point in time the snapshot **156** was taken for various reasons (e.g., data recovery). The remote system **140** may capture any number of snapshots **156** at any intervals (e.g., after a threshold period of time has passed since the last snapshot **156** was captured, whenever a threshold amount of data has been written to the log files **210**, etc.). The snapshots **156**, which represent a state of the block runs **214** at some point in the past, are contrasted against the “live device,” which refers to the state of the block runs **214** of the persistent disk **146P** at the current point in time. That is, the user **12** may query the live device of the persistent disk **146P** to determine a current state of the block runs **214** stored at the persistent disk **146P** or query a snapshot **156** to determine a state of the block runs **214** stored at the persistent disk **146P** for a particular point in time in the past.

(15) To track the locations of the block runs **214** in the log files **210**, each snapshot **156** (and the live device) may include one or more checkpoints **216**, **216a-n** (i.e., checkpoint regions) that include pointers to locations within the log files **210** relevant to the locations of the block runs **214**. In some examples, each checkpoint **216** is a mapping of a logical block address to a location in a respective log file **210**. Optionally, there is a separate log file **210** per segment (e.g., per 32 gigabyte segment). When the remote system **140** receives a read request to retrieve a particular block run **214** (e.g., from the live device or a particular snapshot **156**), the remote system **140** may reference the appropriate checkpoint(s) **216** to determine the one or more log files **210** that include the requested block run **214** and the location (e.g., an offset) within with the log files **210** that include the block run **214** or points to the block run **214**. Some or all of the checkpoints **216** may be cached to volatile memory of the remote system **140** to decrease access times of reads and/or writes. In some examples, the remote system **140** uses other options to reconstruct a device (e.g., tail scanning) when checkpoints **216** are unavailable.

(16) As the log files **210** accumulate data from written block runs **214**, a capacity of the associated persistent disk **146P** decreases. Without garbage collection (i.e., compaction), each persistent disk **146P** will become unavailable for new data once the log files **210** of the persistent disk **146P** reach the end of the device's capacity (e.g., a capacity allotted to the persistent disk **146P** by the remote system and/or a capacity available to the remote system **140**). To this end, the remote system **140** executes a compaction controller **160** that manages garbage collection and/or compaction of the block runs **214** stored within the log files **210**.

(17) The compaction controller **160** obtains a compaction request **20** to compact one or more of the log files **210** of the remote system **140**. The request may come from a user **12**, via another process or module of the remote system **140**, or even generated by the compaction controller **160**. The request may be in response to any number of compaction triggers being satisfied (e.g., a threshold amount of time having passed since a previous compaction, an amount of data written, free capacity of the persistent disks **146P**, deletion of data or a snapshot **156**, etc.). The compaction request **20** may request that all log files **210** be evaluated for compaction or, alternatively, only a subset of the log files **210** be evaluated for compaction. For example, the compaction request **20** may restrict the compaction to log files **210** associated with a particular user **12**, a particular process/module, particular persistent disks **146P** located within a particular geographical region, etc.

(18) For each respective snapshot **156** included by the compaction request **20**, the compaction controller **160** determines, using one or more interval maps **162**, fresh block runs **214**, **214F** of the log files **210** used by the snapshot **156**. Each interval map **162** specifies the fresh block runs **214F** of a respective log file **210** for each snapshot **156**. More specifically, in some implementations, each interval map **162** is a mapping from an offset in a respective log file **210** to data (e.g., metadata) about fresh block runs **214F**. The compaction controller **160** may maintain a separate interval map **162** for each log file **210**. The compaction controller **160** may generate and/or update the interval maps **162**. For example, the compaction controller **160** scans, during a compaction

event, for each snapshot **156**, each log file **210** to generate the interval maps **162**.

(19) The fresh block runs **214F** refer to block runs **214** that are in use. That is, fresh block runs **214F** refer to block runs **214** that have not been modified (i.e., overwritten). This is in contrast to stale block runs **214**, **214S**, which refer to block runs **214** that are no longer in use. For example, a user **12** writes a block run **214** “foo.txt” to a persistent disk **146P**. The remote system **140** writes the block run **214** to an appropriate log file **210** for the persistent disk **146P**. This block run **214**, at this point in time, is a fresh block run **214F**. At a later point in time, the user **12** modifies “foo.txt.” The remote system writes the modified block run **214** to a log file **210**. The new, modified block run **214** is a fresh block run **214F**, however the original block run **214** is now a stale block run **214S**. That is, “foo.txt” now exists twice within the log files **210**, with the first one being stale (because the second one “overwrote” its data) and the second one being fresh.

(20) After determining the fresh block runs **210F** of the log files **210**, the compaction controller **160**, for each respective log file **210**, writes the fresh block runs **214F** of the respective log file **210** to one or more respective new or compacted log files **210**. The fresh block runs **214F** from one respective log file **210**, in some implementations, are spread across multiple new or compacted log files **210**. As explained in more detail below, the new log files **210** include only the fresh block runs **214F** and purposefully exclude any stale block runs **214S**. Thus, the new log files **210** include only fresh block runs **214F** from the original log files **210**. The compaction controller **160** also, for each respective log file **210**, generates a respective per-log diff file **220**, **220a-n**. Each respective per-log diff file **220** maps a location of the written fresh block runs **214F** at the original log file **210** to a location of the written fresh block runs **214F** at the new compacted log file **210**.

(21) The compaction controller **160**, for each respective snapshot **156** (and for the live device), generates a respective new checkpoint **216** for the snapshot **156** based on or using the per-log diff file **220**. That is, instead of reading or scanning the entirety of the snapshot **156**, which may take a substantial amount of memory and other computational resources, the compaction controller **160** leverages the per-log diff file **220** when updating or modifying the original checkpoints **216** or generating the new checkpoints **216** for the snapshots **156**. In some implementations, the compaction controller **160** generates the checkpoints **216** sequentially for the snapshots **156** by starting with the oldest snapshot **156** and finishing with the live device. After the compaction controller **160** generates all of the checkpoints **216**, the new checkpoints **216** are merged into the snapshots **156** and the live device (i.e., the live device (or controlling virtual machine, etc.) is aware of the new mappings), the compaction controller **160** may delete the original, non-compacted log files **210**, thus freeing space for the persistent disk **146P**. In some implementations, the compaction controller **160** automatically deletes any log files **210** that are not referenced by any checkpoint **216**. For example, once the updated checkpoints **216** merge, the original log files **210** are no longer referenced, and the compaction controller **160** may safely delete the unreferenced log files **210**. The checkpoints **216** may merge under any number of conditions. For example, a VM regularly generates new checkpoints **216** (e.g., due to data written to the live device) and merge the checkpoints **216** from the compaction controller **160** simultaneously. The checkpoints **216** may merge during “offline compactions” (e.g., when the controlling VM is offline). The checkpoints **216** may merge immediately upon generation or after a period of time of certain conditions are satisfied.

(22) Referring now to FIG. 2A, a schematic view **200a** includes a simplified example of a persistent disk **146P** represented by two log files **210a-b** (i.e., a “log 1.Math.log” first log file **210a** and a “log 2.Math.log” second log file **210b**). The example also includes two checkpoints **216a-b** that includes a “live device” first checkpoint **216a** and a “snapshot 1” second checkpoint **216b**. The first log file **210a** includes four block runs **214** using six data blocks **212**. The block runs **214** include three fresh block runs **214Fa-c** and one stale block run **214Sa**. The second log file **210b** includes four block runs **214** that use five data blocks **212**. These block runs **214** include three fresh block runs **214Fd-f** and one stale block run **214Sb**. Thus, the two log files **210a-b** store a total of

six fresh block runs **214Fa-f** and two stale block runs **214Sa-b**. The stale block runs **214S** represent an opportunity for compaction or garbage collection.

(23) Both checkpoints **216a-b** include six regions **218**, with each region **218** mapping to one of the six fresh block runs **214F**. In this example, there were no changes to the block runs **214** of the log files **210a-b** in the time between when the snapshot checkpoint **216b** was created and the current live device checkpoint **216a** (which represents the current state of the live device). Accordingly, each of the corresponding regions **218** between the two checkpoints **216a-b** map to the same fresh block run **214F**.

(24) Referring now to FIG. 2B, a schematic view **200b** continues the example of FIG. 2A. Here, two additional data blocks **212** have been written to the second log file **210b** such that the first log file **210a** still has six data blocks **212** of data while the second log file **210b** now has seven data blocks **212** of data. Due to the addition of the new data, the live device checkpoint **216a** updates accordingly to include the mapping to the new data of the log files **210**. Due to the update, the mappings between the first checkpoint **216a** (i.e., for the live device, which updated) and the second checkpoint **216b** (i.e., for the snapshot **156**, which did not update) are no longer the same. Specifically, when an area **250A** of the live device checkpoint **216a** mutated (i.e., updated), the snapshot checkpoint **216b** gained an area **250B** that maps to two locations in the log files **210a-b** that the live device checkpoint **216a** does not map to. While in a conventional log-structured volume without snapshots, these two locations (i.e., fresh block runs **214Ff**, **214Fg**) would be marked stale and be compacted, in this example, the snapshot **156** continues to use the fresh block runs **214Ff**, **214Fg** even after the live device updates, thus necessitating that the fresh block runs **214Ff**, **214Fg** stay fresh. As shown in schematic view **200C** of FIG. 2C, after the live device checkpoint **216a** mutates in response to the two new data blocks written to the second log file **210b**, of the thirteen data blocks **212** of the log files **210a-b**, two are stale and may be compacted, two are in use only by the snapshot **156**, and the remaining nine are in use by the live device and/or the snapshot **156**.

(25) Referring now to FIG. 2D, a schematic view **200D** continues the examples of FIGS. 2A-2C with an example of compacting the log files **210a-b**. For clarity, each the individual files of the log files **210** are not labeled. Here, the original log files **210a-b** include two stale block runs **214Sa-b**. To compact these log files **210a-b**, the compaction controller **160**, in this example, generates two new log files **210c-d** (i.e., “log 3.Math.log” third log file **210c** and “log 3.Math.log” fourth log file **210d**) that include all of the fresh block runs **214F** of the original two log files **210a-b** that the live device checkpoint **216a** references. Notably, the compaction controller **160** does not write the stale block runs **210Sa-b** to the new log files **210c-d**. The compaction controller **160** must also maintain the two fresh block runs **214Ff-g** that the live device checkpoint **216a** does not reference but that the snapshot checkpoint **216b** does reference. While the compaction controller **160** may write these fresh block runs **214Ff-g** to the log files **210c-d**, in this example, the compaction controller **160** generates a third new log file **210e** (i.e., “log 5.Math.log” fifth log file **210e**) that includes only the fresh block runs **214Ff-g** not referenced by the live device checkpoint **216a**. By segregating these fresh block runs **214Ff-g** into a separate log file **210e**, future compactations may be simpler, as when the snapshot **156** associated with the snapshot checkpoint **216b** is deleted, no other checkpoints **216** will reference the associated log file **210e**, and the log file **210e** may be deleted without any other compaction required.

(26) While generating the new log files **210c-e**, the compaction controller **160** also generates a corresponding per-log diff file **220a-b** for each of the original log files **210a-b**. Each per-log diff file **220** tracks movement of blocks still referenced by the live device checkpoint **216** or any snapshot checkpoint **216** from the compacted log file **210** to the new destination (e.g., a newly generated log file **210**). In some implementations, each per-diff log file **220** includes an ordered list of tuples (e.g., a tuple include an offset element, a log file name element, and a byte count element). In some examples, each tuple includes {offset, block run} elements. Here, the offset

element refers to the offset of the first data block **212** of the block run **214**, and the block run **214** refers to a location of a contiguous range of data blocks **212**. Each block run **214** may define the data blocks **212** used by the persistent disk **146P**. The block run **214**, in some examples, includes a tuple with <filename, offset, byte_count> elements.

(27) In the example of FIG. 2D, two log files **210a-b** are compacted, and accordingly, the compaction controller **160** generates two per-log diff files **220a-b**. Here, the first per-log diff file **220a** corresponds to the first log file **210a** and the second per-log diff file **220b** corresponds to the second log file **210b**. The first per-log diff file **220a** maps each of the block runs **214** (or block runs) of the first log file **210a** to their destination locations in the new log files **210c-d**. For example, the block run **214Fa** located at data block '0' in the first log file **210a** is now located at data block '3' of the first new log file **210c**. Similarly, the second per-log diff file **220b** maps the moved block runs **214** and/or block runs of the second log file **210b** to the new logs files **210c-d**. For example, the second per-log diff file **220b** maps the fresh block run **214Fd** located at data block '1' of the second log file **210b** to data block '0' of the first new log file **210c**.

(28) Optionally, the compaction controller **160** periodically deletes the per-log diff files **220**. For example, when the log file **210** the per-log diff file **220** is associated with is deleted (i.e., because the live device and snapshots **156** do not reference the log file **210**), the per-diff log file **220** is also deleted.

(29) The per-log diff files **220** summarize the shuffling of data that happens for each log file **210** during compaction. The per-log diff files **220** are relatively small (e.g., less than 10 mebibytes). Use of per-log diff files **220** instead of using one checkpoint **216** per compacted snapshot **156** is advantageous, as the per-log diff files **220** allow the compaction controller **160** to scale compaction to any number of snapshots **156** without requiring similarly scaling memory requirements. When using checkpoints **216** instead of per-log diff files **220**, when one block **212** is referenced by all snapshots **156**, the movement that occurs during the compaction of this block must be duplicated in all of the in-memory maps (i.e., one per snapshot **156**), which greatly increases memory usage.

(30) Referring now to FIG. 3, in some implementations, the compaction controller **160** determines when to run a compaction (e.g., generate a compaction request **20**) and/or a type of compaction to execute (e.g., a standard compaction that only compacts the live device versus a heavy compaction that compacts the live device and at least some snapshots **156**). In some implementations, the compaction controller **160** executes a heavy compaction (i.e., compacts the snapshots **156**) whenever a snapshot **156** is deleted. However, in some scenarios (e.g., when snapshot **156** churn is high), this technique may generate a large number of compactations. Additionally or alternatively, the compaction controller **160** determines a frequency and/or a type of compaction based on an "IS-density" of the live device and/or one or more snapshots **156**. As used herein, the IS-density refers to a ratio of bytes used by the live device or snapshot **156** over a total size of the log file **210**.

Optionally, the compaction controller **160**, during a compaction event, selects the log files **210** eligible for compaction based on the IS-density of the respective log file **210** (i.e., based on a ratio of a sum of a size of the fresh block runs of each respective log file over a total size of each respective log file). In this way, only a subset of available log files **210** may be compacted during any particular compaction event. As described in more detail below, the compaction controller **160**, in some implementations, determines the IS-density using a respective interval map **162**. In some implementations, only "frozen" log files **210** associated with snapshots **156** are eligible for compaction. Frozen log files **210** refer to log files **210** that are restricted from having new data written. That is, the remote system **140** will not write data to frozen log files **210**.

(31) To determine or track the IS-density, in some examples, the compaction controller **160** maintains a snapshot statistics table **300** for each log file **210**. The compaction controller **160** may generate a new snapshot statistics table **300** periodically (e.g., whenever the compaction controller **160** executes a heavy compaction) and otherwise will maintain the snapshot statistics table **300** until a new snapshot statistics table **300** is generated.

(32) In some implementations, the snapshot statistics table **300** stores a number of statistics **302**. For example, the snapshot statistics table **300** stores a span statistic **302**, **302A**; a garbage statistic **302**, **302B**; and a new blocks statistic **302**, **302C**. The span statistic **302A** defines, per respective snapshot **156** (i.e. per “generation”), a number of bytes in the log file **210** that are mapped or referenced by the respective snapshot **156** and each previous snapshot **156** (i.e., snapshots captured at an earlier point in time relative to the respective snapshot **156**). The garbage statistic **302B** defines, per respective snapshot **156**, an amount of garbage bytes (i.e., bytes that are not referenced by any checkpoint **216**) if the respective snapshot **156** and all previous snapshots **156** (i.e., snapshots captured at an earlier point in time relative to the respective snapshot **156**) were deleted. The new block statistic **302C** defines, per snapshot **156**, a number of bytes in the log file **210** that are new relative to the previous snapshot **156**.

(33) Turning to the exemplary snapshot statistics table **300** of FIG. 3, here, the snapshot statistics table **300** includes statistics for five snapshots **156** (i.e., generations 1-5) and for seventeen bytes (or, alternatively, blocks). The first snapshot **156** (i.e., “Gen 1”) includes ten new data bytes (indicated by the cross-hatch) written to the associated log file **210** (i.e., at byte locations 1-5, 8-10, 13, and 14) and seven byte locations are still empty. Accordingly, the span statistic **302A** and the new block statistic **302C** for the first snapshot **156** are ten. The garbage statistic **302B** (which is four in this example), requires knowledge from the next snapshot **156** to determine. The second snapshot includes seven data bytes written to the associated log file (i.e., at byte locations 3-6, 9, 10, and 13). Four of the data bytes used by the first snapshot **156** are no longer used by the second snapshot, and therefore the garbage statistic **302B** for the first snapshot **156** is four. The second snapshot **156** uses one additional data byte not used by the first snapshot **156**, so the span statistic **302A** for the second snapshot (i.e., “Gen 2”) grows to eleven while the new block statistic **302C** is set to one. A third snapshot **156** includes data bytes at byte locations 4, 5, 9, 11, 12, 15, and 16. The four new bytes added (i.e., at byte locations 11, 12, 15, and 16) set the span statistic **302A** for the third snapshot **156** to fifteen and the new block statistic **302C** to four. The snapshot statistics table **300** continues in this fashion for each snapshot **156** that references the associated log file **210**. In some examples, the compaction controller **160** builds each snapshot statistics table **300** when performing a compaction (e.g., a heavy compaction) using the corresponding interval maps **162** and checkpoints **216**.

(34) At the beginning of a compaction, the compaction controller **160** may determine a number of checkpoints **216** and determine which checkpoints **216** have been deleted. However, storing all possibilities in the snapshot statistics table **300** generally requires significant resources as the number of snapshots **156** grows (i.e., in order to have data in case any subset of snapshots **156** are deleted). Instead, in some examples, the snapshot statistics table **300** is ideal for rolling schedules of snapshots **156** where the older snapshots **156** are deleted first. For example, if all snapshots **156** with generation a lower than N are removed, the compaction controller **160** determines a minimum number of garbage bytes based on the garbage statistic **302B** from the N-1 snapshot **156**. This is a minimum, as there may be potentially more garbage bytes from live device activity or from other deleted snapshots **156** with a generation larger than N. Accordingly, the compaction controller **160** may determine an upper bound of the IS-density of a respective log with

$$(35) \quad \text{ISD} = \frac{L - \text{garbage}[N-1]}{L} \quad (1)$$

(36) Here, ISD refers to the IS-density, L refers to a size (e.g., in bytes) of the respective log file **210**, and garbage[N-1] refers to the garbage statistic **302B** of the previous snapshot **156**.

(37) Using the new block statistic **302C** from the snapshot statistics table **300**, the compaction controller **160**, in some examples, determines major changes in bytes mapped by different snapshots **156**. In the example snapshot statistics table **300** of FIG. 3, it is clear that, while the third snapshot **156** (i.e., “Gen 3”) reuses a lot of blocks used by previous snapshots **156**, the snapshot **156** also introduces a lot of fresh data. When two adjacent snapshots **156** both introduce a lot of

fresh data, deletion of the older snapshot **156** may indicate an appropriate time to execute a heavy compaction. The new block statistic **302C** can correctly be summed for adjacent snapshots. For example, when generations [X, Y] are deleted and their summed new bytes (determined from the new block statistic **302C**) is significant (e.g., above a threshold amount), while the new block statistic **302C** for the snapshot **156** [Y+1] is also significant, then a compaction may also yield significant byte savings.

(38) FIG. 4 is a flowchart of an exemplary arrangement of operations for a method **400** of garbage collection in log-based block devices with snapshots. The computer-implemented method **400**, when executed by data processing hardware **144**, causes the data processing hardware **144** to perform operations. The method **400**, at operation **402**, includes obtaining a request **20** to compact a plurality of log files **210** of a log-structured volume **146P**. Each log file **210** of the plurality of log files **210** may include fresh block runs **214F** in use by the log-structured volume **146P** and stale block runs **146S** no longer in use by the log-structured volume **146P**. The log-structured volume **146P** includes a plurality of snapshots **156**. Each snapshot **156** of the plurality of snapshots **156** is representative of a state of the log-structured volume **146P** at a different point in time.

(39) At operation **404**, the method **400** includes, for each respective snapshot **156** of the plurality of snapshots **156** of the log-structured volume **146P**, determining the fresh block runs **214F** of the plurality of log files **210** used by the respective snapshot **156**. Optionally, the method **400** includes using a plurality of interval maps **162** to determine the fresh block runs **214F**. Each respective interval map **162** of the plurality of interval maps **162** specifies the fresh block runs **214F** of a respective log file **210** for each snapshot **156** of the plurality of snapshots **156**. For each respective log file **210** of the plurality of log files **210**, the method **400**, at operation **406**, includes writing the fresh block runs **216F** of the respective log file **210** to a respective compacted log file **210** and generating a respective per-log diff file **220** mapping a location of the written fresh block runs **214F** at the respective log file **210** to a location of the written fresh block runs **214F** at the respective compacted log file **210**.

(40) At operation **408**, the method **400** includes, for each respective snapshot **156** of the plurality of snapshots **156** of the log-structured volume **146P**, generating a respective checkpoint **216** based on respective per-log diff files **220**. At operation **410**, the method **400** includes deleting each respective log file **210** of the plurality of log files **210**.

(41) FIG. 5 is a schematic view of an example computing device **500** that may be used to implement the systems and methods described in this document. The computing device **500** is intended to represent various forms of digital computers, such as laptops, desktops, workstations, personal digital assistants, servers, blade servers, mainframes, and other appropriate computers. The components shown here, their connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the inventions described and/or claimed in this document.

(42) The computing device **500** includes a processor **510**, memory **520**, a storage device **530**, a high-speed interface/controller **540** connecting to the memory **520** and high-speed expansion ports **550**, and a low speed interface/controller **560** connecting to a low speed bus **570** and a storage device **530**. Each of the components **510**, **520**, **530**, **540**, **550**, and **560**, are interconnected using various busses, and may be mounted on a common motherboard or in other manners as appropriate. The processor **510** can process instructions for execution within the computing device **500**, including instructions stored in the memory **520** or on the storage device **530** to display graphical information for a graphical user interface (GUI) on an external input/output device, such as display **580** coupled to high speed interface **540**. In other implementations, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. Also, multiple computing devices **500** may be connected, with each device providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).

(43) The memory **520** stores information non-transitorily within the computing device **500**. The memory **520** may be a computer-readable medium, a volatile memory unit(s), or non-volatile memory unit(s). The non-transitory memory **520** may be physical devices used to store programs (e.g., sequences of instructions) or data (e.g., program state information) on a temporary or permanent basis for use by the computing device **500**. Examples of non-volatile memory include, but are not limited to, flash memory and read-only memory (ROM)/programmable read-only memory (PROM)/erasable programmable read-only memory (EPROM)/electronically erasable programmable read-only memory (EEPROM) (e.g., typically used for firmware, such as boot programs). Examples of volatile memory include, but are not limited to, random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), phase change memory (PCM) as well as disks or tapes.

(44) The storage device **530** is capable of providing mass storage for the computing device **500**. In some implementations, the storage device **530** is a computer-readable medium. In various different implementations, the storage device **530** may be a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid state memory device, or an array of devices, including devices in a storage area network or other configurations. In additional implementations, a computer program product is tangibly embodied in an information carrier. The computer program product contains instructions that, when executed, perform one or more methods, such as those described above. The information carrier is a computer- or machine-readable medium, such as the memory **520**, the storage device **530**, or memory on processor **510**.

(45) The high speed controller **540** manages bandwidth-intensive operations for the computing device **500**, while the low speed controller **560** manages lower bandwidth-intensive operations. Such allocation of duties is exemplary only. In some implementations, the high-speed controller **540** is coupled to the memory **520**, the display **580** (e.g., through a graphics processor or accelerator), and to the high-speed expansion ports **550**, which may accept various expansion cards (not shown). In some implementations, the low-speed controller **560** is coupled to the storage device **530** and a low-speed expansion port **590**. The low-speed expansion port **590**, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet), may be coupled to one or more input/output devices, such as a keyboard, a pointing device, a scanner, or a networking device such as a switch or router, e.g., through a network adapter.

(46) The computing device **500** may be implemented in a number of different forms, as shown in the figure. For example, it may be implemented as a standard server **500a** or multiple times in a group of such servers **500a**, as a laptop computer **500b**, or as part of a rack server system **500c**.

(47) Various implementations of the systems and techniques described herein can be realized in digital electronic and/or optical circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

(48) A software application (i.e., a software resource) may refer to computer software that causes a computing device to perform a task. In some examples, a software application may be referred to as an “application,” an “app,” or a “program.” Example applications include, but are not limited to, system diagnostic applications, system management applications, system maintenance applications, word processing applications, spreadsheet applications, messaging applications, media streaming applications, social networking applications, and gaming applications.

(49) These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine

language. As used herein, the terms “machine-readable medium” and “computer-readable medium” refer to any computer program product, non-transitory computer readable medium, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor.

(50) The processes and logic flows described in this specification can be performed by one or more programmable processors, also referred to as data processing hardware, executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

(51) To provide for interaction with a user, one or more aspects of the disclosure can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube), LCD (liquid crystal display) monitor, or touch screen for displaying information to the user and optionally a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser.

(52) A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. Accordingly, other implementations are within the scope of the following claims.

Claims

1. A computer-implemented method executed by data processing hardware that causes the data processing hardware to perform operations comprising: obtaining a request to compact a plurality of first log files of a log-structured volume, the log-structured volume comprising a plurality of snapshots, each snapshot of the plurality of snapshots representative of a state of the log-structured volume at a different point in time; for each respective snapshot of the plurality of snapshots of the log-structured volume, determining fresh block runs of the plurality of first log files in use by the respective snapshot of the log-structured volume, the fresh block runs representing contiguous blocks of data that have not been overwritten; for each respective first log file of the plurality of

first log files: writing the fresh block runs of the respective first log file to one or more respective second log files different from the plurality of first log files; and generating a respective per-log diff file mapping a location of the written fresh block runs at the respective first log file to a location of the written fresh block runs at the one or more respective second log files; for each respective snapshot of the plurality of snapshots of the log-structured volume, generating a respective checkpoint based on respective per-log diff files; and deleting each respective first log file of the plurality of first log files.

2. The method of claim 1, wherein: determining the fresh block runs comprises using a plurality of interval maps, each respective interval map of the plurality of interval maps specifying the fresh block runs of a respective first log file for each snapshot of the plurality of snapshots; and the operations further comprise generating, using the plurality of first log files and the plurality of snapshots, the plurality of interval maps.

3. The method of claim 2, wherein generating the plurality of interval maps comprises scanning, for each snapshot of the plurality of snapshots, each first log file of the plurality of first log files.

4. The method of claim 1, wherein each fresh block run and each stale block run comprises a tuple comprising a log file name, an offset, and a byte count.

5. The method of claim 1, wherein the operations further comprise selecting the plurality of first log files from a set of first log files based on a ratio of a sum of a size of the fresh block runs of each respective first log file over a total size of each respective first log file.

6. The method of claim 5, wherein the operations further comprise, for each respective first log file of the set of first log files, determining the ratio for each respective first log file using a respective interval map specifying the fresh block runs of a respective first log file for each snapshot of the plurality of snapshots.

7. The method of claim 1, wherein each first log file of the plurality of first log files is frozen.

8. The method of claim 1, wherein obtaining the request to compact the plurality of first log files comprises determining a threshold amount of time has passed since a previous compaction.

9. The method of claim 1, wherein each respective per-log diff file comprises an ordered list of tuples, each tuple representing a respective fresh block run and comprising an offset, a log file name, and a byte count.

10. The method of claim 1, wherein the operations further comprise, for each deleted respective first log file, deleting the respective per-log diff file.

11. The method of claim 1, wherein each first log file of the plurality of first log files comprises the fresh block runs representing respective contiguous blocks of data in use by the log-structured volume and stale block runs representing respective contiguous blocks of data no longer in use by the log-structured volume.

12. A system comprising: data processing hardware; and memory hardware in communication with the data processing hardware, the memory hardware storing instructions that when executed on the data processing hardware cause the data processing hardware to perform operations comprising: obtaining a request to compact a plurality of first log files of a log-structured volume, the log-structured volume comprising a plurality of snapshots, each snapshot of the plurality of snapshots representative of a state of the log-structured volume at a different point in time; for each respective snapshot of the plurality of snapshots of the log-structured volume, determining fresh block runs of the plurality of first log files in use by the respective snapshot of the log-structured volume, the fresh block runs representing contiguous blocks of data that have not been overwritten; for each respective first log file of the plurality of first log files: writing the fresh block runs of the respective first log file to one or more respective second log files different from the plurality of first log files; and generating a respective per-log diff file mapping a location of the written fresh block runs at the respective first log file to a location of the written fresh block runs at the one or more respective second log files; for each respective snapshot of the plurality of snapshots of the log-structured volume, generating a respective checkpoint based on respective per-log diff files; and

deleting each respective first log file of the plurality of first log files.

13. The system of claim 12, wherein: determining the fresh block runs comprises using a plurality of interval maps, each respective interval map of the plurality of interval maps specifying the fresh block runs of a respective first log file for each snapshot of the plurality of snapshots; and the operations further comprise generating, using the plurality of first log files and the plurality of snapshots, the plurality of interval maps.

14. The system of claim 13, wherein generating the plurality of interval maps comprises scanning, for each snapshot of the plurality of snapshots, each first log file of the plurality of first log files.

15. The system of claim 12, wherein each fresh block run and each stale block run comprises a tuple comprising a log file name, an offset, and a byte count.

16. The system of claim 12, wherein the operations further comprise selecting the plurality of first log files from a set of first log files based on a ratio of a sum of a size of the fresh block runs of each respective first log file over a total size of each respective first log file.

17. The system of claim 16, wherein the operations further comprise, for each respective first log file of the set of first log files, determining the ratio for each respective first log file using a respective interval map specifying the fresh block runs of a respective first log file for each snapshot of the plurality of snapshots.

18. The system of claim 12, wherein each first log file of the plurality of first log files is frozen.

19. The system of claim 12, wherein obtaining the request to compact the plurality of first log files comprises determining a threshold amount of time has passed since a previous compaction.

20. The system of claim 12, wherein each respective per-log diff file comprises an ordered list of tuples, each tuple representing a respective fresh block run and comprising an offset, a log file name, and a byte count.

21. The system of claim 12, wherein the operations further comprise, for each deleted respective first log file, deleting the respective per-log diff file.

22. The system of claim 12, wherein each first log file of the plurality of first log files comprises the fresh block runs representing respective contiguous blocks of data in use by the log-structured volume and stale block runs representing respective contiguous blocks of data no longer in use by the log-structured volume.
